

In the textbook [Godsil], [Mieghem], they define the symmetric Laplacian as follows:

$$(1.b) \quad \boxed{L^{\text{sym}} = D - A} \quad \text{or,} \quad \boxed{L_{ij}^{\text{sym}} = k_i \delta_{ij} - a_{ij}} \quad \text{--- (1a)}$$

$D = \text{diag}(k_1, k_2, \dots, k_N)$
 A : Adjacency matrix.

Now, from random walk on a graph, we obtain (as derived in my last note)

$$\boxed{L_{ij}^{\text{RW}} = \delta_{ij} - \frac{a_{ij}}{k_i}} \quad \text{--- (2a)}$$

or,

$$\boxed{L^{\text{RW}} = I - D^{-1}A} \quad \text{--- (2b)}$$

On the other hand, defined as

$$L_{ij}^{\text{SN}} = \delta_{ij} - \frac{a_{ij}}{\sqrt{k_i k_j}} \quad \text{--- (3a)}$$

or,

$$\boxed{L^{\text{SN}} = I - D^{-1/2} A D^{-1/2}} \quad \text{--- (3b)}$$

Thus, we can see from Eqs (2b) and (3b), that L^{RW} and L^{SN} are related via the following similarity transformation:

$$\begin{aligned} L^{\text{RW}} &= I - D^{-1} \left[D^{1/2} (I - L^{\text{SN}}) D^{1/2} \right] \\ &= I - D^{-1/2} (I - L^{\text{SN}}) D^{1/2} \quad \text{--- (4)} \end{aligned}$$

Now, note that, the eqn which we obtain for conservation of probability :

$$\sum_i \langle i | L^T = 0 \quad \dots (\text{Eq. 5}) \quad \text{holds for}$$

non-symmetric L^{RW} and also for symmetric L^{sym} , as defined in [Miegrem, Godsil].

Since, arriving at (Eq. 5) (the above one) for L^{SN} is not straightforward, we may look at how its eigenvectors are related to L^{RW} . Anyway, since L^{RW} , L^{sym} and L^{SN} are all similar, ^{that of} they will have the same set of eigenvalues.

Eigenvectors of L^{SN} :

~~Right ei~~ Let, $|v^R\rangle$: Right eigenvector of L^{RW} corresponding to eigenvalue λ .

Then,

$$L^{RW} |v^R\rangle = \lambda |v^R\rangle \quad \dots \textcircled{6}$$

$$\Rightarrow [I - D^{-1/2} (I - L^{SN}) D^{1/2}] |v^R\rangle = \lambda |v^R\rangle \quad [\text{Using Eq. 4}]$$

(After simplification)

$$\Rightarrow \underbrace{L^{SN} D^{1/2} |v^R\rangle}_{\text{}} = \underbrace{\lambda D^{1/2} |v^R\rangle}_{\text{}} \quad \dots \textcircled{7}$$

thus, $|v_{SN}^R\rangle = D^{1/2} |v^R\rangle$ is the eigenvector of L^{SN} corresponding to eigenvalue λ (corresponding to right eigenvector $|v^R\rangle$ of L^{RW})

Similarly, one can easily show,

$$\langle v_{SN}^L | = D^{-1/2} \langle v^L |$$

In fact, L^{SN} is symmetric, so, we can simply write

$$\begin{aligned} |v_{SN}\rangle &= D^{1/2} |v_R\rangle \\ \langle v_{SN}| &= \cancel{\langle v^L|} D^{-1/2} \end{aligned} \quad \text{--- (8)}$$

The relations in Eq. (11) and (12) of our paper are for

$|v^R\rangle$ and $\langle v^L|$ of L^{RW} .

We can then simply substitute those $|v^R\rangle$ and $\langle v^L|$ with $|v_{SN}\rangle$ and $\langle v_{SN}|$ (following Eq. (8) (above eqn)). to go to the basis of L^{SN} .

□ In fact, the normalization condition now yields

$$\sum_i \sqrt{\frac{k_i}{k_j}} \langle i | L^{SN} = 0$$

Derivation:

$$\begin{aligned} \sum_i \langle i | (L^{RW})^T | j \rangle &= 0 \\ \Rightarrow \sum_i \langle i | D^{1/2} L^{SN} D^{-1/2} | j \rangle &= 0 \\ \Rightarrow \sum_i \sqrt{\frac{k_i}{k_j}} \langle i | L^{SN} | j \rangle &= 0 \end{aligned}$$